John Leiga

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U.S. Citizen • Eligible to work on export-controlled programs

Education

B.S. Mechanical Engineering 2017 – 2021
University of California, Merced

Experience

Nestle Purina PetCare Company Bloomfield, MO — Ft. Dodge, IA — St. Louis, MO
Design Engineer Feb 2025 – Present

- Designed extrusion dies, production tooling, and equipment components in Autodesk Inventor based on factory needs, production feedback, and manufacturing constraints.
- Owned unit operation specifications that defined approved equipment, operating standards, and process requirements for factory systems.
- Translated factory feedback into equipment requirements, tooling updates, operating standards, and design changes to improve reliability and process consistency.
- Validated process equipment during factory startup through installation readiness checks, startup support, production ramp troubleshooting, and operational handoff.

Management Development Associate - Engineering May 2023 – Feb 2025

- Optimized a 14-furnace perlite expansion process by tuning control loops in sequence, debugging process instability, and adding density-based feedback to improve product consistency.
- Diagnosed a recurring perlite process bottleneck by tracing product backup and shutdowns to an undersized rotary airlock in the main product stream, creating the basis for equipment correction and process stabilization.
- Structured perlite DOE trials using variable screening, interaction testing, high-resolution data capture, and Minitab/Excel analysis to identify run parameters for different ore types.
- Developed and deployed a zero-waste automated startup sequence, saving $200K+ annually across Missouri and Virginia facilities.

Tavis Corporation Mariposa, CA
Project Engineer Sep 2021 – May 2023

- Designed aerospace pressure transducer components and assemblies in SolidWorks, applying stress analysis and producing engineering drawings, assembly drawings, and BOMs to support prototype build, testing, manufacturing documentation, and Manufacturing Readiness Review preparation.
- Applied first-principles mechanical analysis to design and prototype-validate a differential pressure transducer architecture, reducing mechanical component cost by approximately 50% and total hardware cost by approximately 30%.
- Interpreted long-form aerospace specifications to define pressure range, media compatibility, environmental, test, documentation, and qualification requirements for custom hardware.
- Developed qualification test procedures to validate pressure transducer survival and operation under thermal, vibration, shock, and overpressure requirements.
- Led internal design reviews and customer-facing Critical Design Reviews, aligning requirements, design tradeoffs, test readiness, qualification data, compliance documentation, open actions, and technical deliverables.
- Managed aerospace hardware programs through qualification and manufacturing integration, owning requirements, schedules, customer communication, and technical deliverables.
- Led failure analysis, documentation updates, and component obsolescence reviews to support production continuity and customer program execution.

Skills

Mechanical Design: SolidWorks, Autodesk Inventor, precision mechanical design, aerospace pressure transducer hardware, electromechanical sensor assemblies, DFM, engineering drawings, assembly drawings, BOMs

Requirements and Verification: requirements interpretation, aerospace specification review, qualification planning, test procedures, compliance documentation, Critical Design Reviews, customer-facing design reviews

Manufacturing and Product Support: production support, manufacturing documentation, Manufacturing Readiness Reviews, manufacturing integration, startup validation, root cause analysis, failure analysis

Tools: SolidWorks, Autodesk Inventor, Microsoft Excel, Minitab, Python, MATLAB, Git

Independent Projects

Python ML Research System

- Built a modular Python research platform for equities and options strategy development, using artifact-based pipeline stages for feature engineering, labeling, LightGBM model training, backtesting, model explainability, and live inference.